

Javascript

Operators, type coercion, conditional statement ,for loop

Theory Notes

1. Operators

- **Arithmetic** → +, -, *, /, %.
- **Relational** → >, <, >=, <=, ==, ===, !=, !==.
- **Logical** → &&, ||, !.
- **Bitwise** → &, |, ^, >>, <<, ~.
- **Assignment** → =, +=, -=, *=, /=, %=.
- **Ternary** → (condition) ? expr1 : expr2.

2. Type Coercion

- **Implicit** → JavaScript automatically converts types.
 - `1 + "3" → "13" (string).`
 - `2 * "3" → 6 (number).`
- **Explicit** → manual conversion using `String()`, `Number()`, `parseInt()`.

3. Conditional Statements

- `if`, `if-else`, `nested if`, `else-if ladder`, `switch`.
- Used for decision-making based on conditions.

4. Loops

- **for** → repeat with known iterations.
- **while** → repeat until condition is false.
- **do-while** → executes at least once, then checks condition.

Coding Round Questions

1. Write a program to check if a number is even or odd using the modulus operator.
2. Demonstrate the difference between `==` and `===` with examples.
3. Use a ternary operator to check if a student has passed (marks ≥ 35).
4. Write a program to calculate the sum of numbers from 1 to 10 using a `for` loop.
5. Demonstrate type coercion with `1 + "2"` and `2 * "3"`.
6. Write a program using `switch` to print grades based on marks.

Challenge

Problem: Student Grading System

Requirements:

- Input: student's marks.
- Output: grade (O, A, B, Fail).
- Use:
 - Relational operators to check ranges.
 - Ternary operator for pass/fail.
 - Switch statement for exact marks.
- Bonus: Loop through an array of student marks and print grades for each.

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Operators</title>
  </head>
  <body>
    <h1>Operators</h1>

    <!-- binary operators -> arithmetic, relational, logical, bitwise,
assignment -->

    <!-- relational operator -> > , < , >= , <= , == , === , != , !== -->

    <script>
      let a = 10;
      let b = 10;

      //    10 > 20

      console.log(b > a);

      console.log(a < b);

      console.log(a >= b);

      console.log(a <= b);

      console.log(a == b); // check only value

      console.log(a === b); // check for value and check for data type

      console.log(a != b);

      console.log(a !== b);

      // logical operator  && , || , !
```

```
console.log(true && true);
console.log(false && true);
console.log(true && false);
console.log(false && false);

// true, true -> true
// false, true -> false
// true, false -> false
// false, false -> false

//OR

// true, true -> true
// false, true -> true
// true, false -> true
// false, false -> false
console.log(true || true);
console.log(false || true);
console.log(true || false);
console.log(false || false);

// !
// true -> false
// false -> true

console.log(!true);
console.log(!false);

// assignment operator
// +=, -=, *=, /=, %=

let salary = 50000;

salary += 10000;

console.log(salary);

// bitwise operator
// & | ^ >> << ~

let first = 5;
let second = 3;

console.log(first & second);

console.log(first | second);
console.log(first ^ second);
console.log(first >> 1);
```

```
console.log(first << 1);

// ternary operator

var marks = 7;
// (condition) ? expression1 : expression2;

var result1 = marks > 35 ? "Pass" : "Fail";
console.log(result1);

// type coercion

// implicit and explicit

let result = 1 + "3";

console.log(typeof result);

console.log(1 - "2");

console.log(2 * "3");

var data = String(123);
var data1 = Number("123");

var data2 = parseInt("123px");

console.log(typeof data);
console.log(typeof data1);
console.log(data1);
console.log(data2);

// conditional statements
// if, if-else, nested if, if-else-ladder, switch

let age = 18;
let voterId = false;

if (false) {
  console.log("Statement is true");
}

if (age >= 18) {
  console.log("You are eligible to vote");
} else {
  console.log("You are not eligible for vote");
}
```

```
if (age >= 18) {
  if (voterId) {
    console.log("you are eligible for vote");
  } else {
    console.log("You don't have voter id ");
  }
} else {
  console.log("You are under 18");
}
```

```
// else if ladder
```

```
var markss = 2000;
```

```
if (markss >= 90) {
  console.log("O");
} else if (markss >= 80) {
  console.log("A");
} else if (markss >= 70) {
  console.log("B");
} else if (marks < 30) {
  console.log("Fail");
} else {
  console.log("Condition does not meet");
}
```

```
// check for even number
```

```
let num = 9;
```

```
if (num % 2 == 0) {
  console.log("Even number");
} else {
  console.log("odd number");
}
```

```
// switch
```

```
switch (markss) {
  case 90:
    console.log("O");
    break;

  case 80:
    console.log("A");
    break;

  default:
```

```
        console.log("no mark matches");
        break;
    }

    // loops
    // for, while, do-while loop

    for(let i=1 ; i<=10; i+=2){
        console.log(i);
    }
</script>
</body>
</html>
```